1.) Identify your problem statement

Machine learning →Supervised learning →Regression.

2.) Tell basic info about the dataset (Total number of rows, columns)

6-Rows and 1339 Columns.

3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data).

One-Hot Encoding (Dummy Encoding)

1.multiple linear regression r^2=0.78

2.Support vector machine

S.No	Hyperparameter (c)	Linear	RBF	Ploy	Sigmoid
1	0.01	-0.09	-0.09	-0.09	-0.098
2	0.1	-0.15	-0.09	-0.09	-0.098
3	0.5	-0.15	-0.09	-0.08	-0.098
4	1	-0.14	-0.09	-0.07	-0.098
5	100	0.52	-0.15	-0.13	-0.12
6	1000000	0.76	0.78	0.85	-102

3.Decision tree

S.No	*Criterion*	*Max feature*	*Splitter*	*R^2 Value*
1	*Squared error*	*sqrt*	*Best*	*0.74*
2		*sqrt*	*Random*	*0.71*
3		*Log2*	*Best*	*0.66*
4		*Log2*	*Random*	*0.74*
5		*Auto*	*Best*	*0.70*
6		*Auto*	*Random*	*0.67*
7	*Absolute error*	*sqrt*	*Best*	*0.74*
8		*sqrt*	*Random*	*0.70*
9		*Log2*	*Best*	*0.70*
10		*Log2*	*Random*	*0.73*

11		Auto	Best	0.65
12		Auto	Random	0.74
13	Friedman mse	sqrt	Best	0.60
14		sqrt	Random	0.68
15		Log2	Best	0.63
16		Log2	Random	0.71
17		Auto	Best	0.69
18		Auto	Random	0.71

4.Random forest

S.No	Criterion	Max feature	N_estimators	R^2 Value
1	Squared error	sqrt	10	0.87
2		sqrt	100	0.85
3		Log2	10	0.83
4		Log2	100	0.86
5		Auto	10	0.84
6		Auto	100	0.85
7	Absolute error	sqrt	10	0.85
8		sqrt	100	0.87
9		Log2	10	0.85
10		Log2	100	0.87
11		Auto	10	0.84
12		Auto	100	0.85

Best model is Random forest (Squard error ,sqrt ,N_estimators *is 10)*R^2 *Value=0.87*